Sprint Ceremony Document: Sprint 4

Project Name: ALT-F4
Sprint Window: November 1 - November 7, 2026
Overall Completion: 95/168 * 100 = **56.55%**
Sprint 4 Completion: 24/24 * 100 = **100%**

Sprint Planning & Progress Update

Completed Tasks:

1. Queue Workflow Definition (CQ-S4-4.1)

- Online and Walk-in Queue Planning: Defined the queue workflow for patients with online appointments and walk-in patients, establishing the expected process for entering and being managed in the clinic queue.
- Queue Behavior Planning: Established the initial workflow considerations for active and closed queues, ticket numbering, queue position, estimated waiting time, priority, timestamps, tracking tokens, and duplicate-entry prevention.

2. Queue Entry API & Ticket Generation (CQ-S4-4.2)

- Input Validation: Implemented validation for patient names, optional visit reasons, and patient ID mapping to ensure valid queue entries.
- Sequential Ticket Generation: Implemented automatic ticket number generation using the `A-001` format, with new entries assigned an initial `WAITING` status.
- Secure Tracking Tokens: Generated hashed tracking tokens to support secure queue lookups for patients without exposing sensitive information.
- Dynamic Queue Position & Wait Estimation: Implemented queue position calculation and estimated waiting time based on the number of active entries and fallback service duration.
- Online Queue Entry API: Implemented `POST /api/v1/queues/:queueId/online` and integrated it with the Patient Dashboard, allowing patients to obtain queue tickets through the web application.
- Queue Availability Validation: Added handling for closed and nonexistent queues through descriptive `QUEUE_CLOSED` and `QUEUE_NOT_FOUND` error responses.

3. Queue Status API, Integrity & Real-Time Synchronization (CQ-S4-4.3)

- Real-Time Queue Status: Implemented `GET /api/v1/queues/:queueId/status`, providing the currently serving ticket, counter or room designation, total waiting patients,

queue category breakdown, completed appointments for the day, and average waiting time.

- Duplicate Entry Prevention: Enforced `DUPLICATE_ACTIVE_ENTRY` validation to prevent patients from holding multiple active queue tickets within the same daily queue session.
- Database Integrity: Integrated Prisma unique-constraint conflict handling through `P2002` error mapping to maintain queue entry integrity.
- Secure Patient Queue Tracking: Implemented `GET /api/v1/queues/:queueId/my-entry`, allowing authenticated patients to retrieve their own queue position, status, and estimated waiting time without exposing other patients' records or personal information.
- Live Queue Synchronization: Connected Staff Console counter progression and countdown timers to queue operations, while integrating the Patient Dashboard's Now Serving metrics with Server-Sent Events (`queue:called` and `queue:updated`).
- Transaction Optimization: Increased Prisma interactive transaction timeouts to 15 seconds to accommodate remote database latency and restricted database locks to atomic write operations.

4. Patient Queue Interface & UI/UX Redesign (CQ-S4-4.4)

- Staff, Patient & Authentication Redesign: Redesigned the staff dashboard, patient interface, and login/register screens with an updated brand palette, logo, collapsible sidebar, and statistical cards.
- Patient Virtual Queue Screen: Developed a dedicated queue page featuring a ticket information card and queue summary card, accessible through the sidebar.
- Live Queue Status Integration: Connected the queue summary to the backend status API, displaying updated queue information with loading, closed, and error states.
- Queue Monitoring Indicators: Added a Refresh button and Live/Offline indicators to communicate the current connection and synchronization state.
- Empty & Feedback States: Implemented a "You are not in the queue" empty state, next-in-line and duplicate-entry banners, and appropriate status label mappings.

5. Patient Dashboard Queue Integration & Automated Entry (CQ-S4-4.5)

- Automatic Queue Entry: Integrated appointment booking with automatic generation of a sequential queue ticket (`A-XXX`) in `WAITING` status for the current day's queue, eliminating the need for a separate manual Join Queue action.
- Accurate Waiting Time Calculation: Corrected the estimated waiting time formula to calculate based on `patientsAhead × 15 minutes` instead of multiplying the patient's queue position by the service duration.

- Unified Appointment & Queue Cancellation: Replaced the separate Cancel Ticket action with a unified Cancel Appointment operation that cancels both the appointment and its associated queue ticket in the database.
- Appointment Requirement Validation: Prevented patients without confirmed appointments from entering the queue and provided a Book to Queue option to direct them to the appointment booking process.
- Live Backend Integration: Connected the `/patient/queue` interface to the live `getMyQueueEntry` and `trackByToken` backend services, replacing previously used mock queue data.

Planned Tasks for Next Sprint (Sprint 5):

Task ID	Task Title	Description
CQ-S5-5.1	Connect the staff dashboard to appointments and active queue	Show upcoming appointments, waiting patients, currently serving ticket/room and operational counts with appropriate empty/no-queue states.
CQ-S5-5.2	Implement call-next and queue status APIs	Implement call-next behavior, room/provider information, timestamps and valid queue entry lifecycle transitions.
CQ-S5-5.3	Build clinic staff queue-operation controls	Build staff controls for Call Next, Start Consultation, Complete and Mark No-Show, plus any explicitly approved skip/return action.
CQ-S5-5.4	Synchronize queue, appointment and consultation state	Keep queue/appointment statuses, timestamps and operational metrics synchronized; ensure patient-facing views expose only appropriate status information.
CQ-S5-5.5	Test queue operations and state transitions	Test ordered calls, empty queues, invalid room/provider details, concurrent staff actions and all supported status transitions.
CQ-S5-5.6	Demonstrate the full clinic staff operational cycle	Demonstrate a patient from queue entry through call, consultation and outcome; triage findings and prepare the notification/audit event list.

Sprint Retrospective

What Went Well:

- **Queue Workflow Planning & Entry Management (CQ-S4-4.1 & CQ-S4-4.2):** Established the queue workflow for online and walk-in patients and implemented validated queue entry mechanisms. The system now generates sequential ticket

numbers, assigns initial waiting statuses, calculates queue positions, and provides estimated waiting times.

- **Secure Queue Tracking & Data Integrity (CQ-S4-4.2 & CQ-S4-4.3):** Implemented hashed tracking tokens, authenticated patient queue retrieval, and duplicate-entry prevention. These mechanisms help protect patient information while maintaining the integrity of active queue records.
- **Real-Time Queue Monitoring (CQ-S4-4.3 & CQ-S4-4.4):** Integrated queue status APIs with Server-Sent Events to synchronize Now Serving information and queue updates between the backend, Staff Console, and Patient Dashboard.
- **Patient Virtual Queue Interface (CQ-S4-4.4):** Delivered a dedicated patient queue screen with ticket information, queue position, estimated waiting time, current serving details, and appropriate loading, empty, closed, and error states.
- **Automated Queue Entry & Appointment Integration (CQ-S4-4.5):** Integrated automatic queue ticket generation with appointment booking, allowing patients to enter the queue without an additional manual action. Appointment cancellation also automatically updates the associated queue ticket.
- **Queue Calculation & Database Optimization (CQ-S4-4.3 & CQ-S4-4.5):** Improved waiting time accuracy by basing estimates on the number of patients ahead. Optimized Prisma transactions and database locking to support more reliable operations with the remote cloud database.

What Could Be Improved / Challenges Encountered:

- **Completion of the Staff Queue Operational Workflow:**
    - Challenge: The current sprint established queue entry, monitoring, and patient-facing tracking. However, the complete staff-side operational process, including calling patients, starting consultations, completing visits, and handling no-shows, still needs to be integrated.
    - Resolution: Established the queue status APIs and real-time synchronization mechanisms needed to support staff queue operations.
    - Future Improvement: Complete the staff operational controls and integrate every supported queue transition with appointment and consultation records in the next sprint.
- **Queue, Appointment & Consultation State Synchronization:**
    - Challenge: Queue tickets and appointments are interconnected. Changes in one record must be reflected correctly in the other parts of the system to prevent inconsistent statuses and operational records.
    - Resolution: Implemented automatic queue entry during appointment booking and unified cancellation of appointments and their associated queue tickets.
    - Future Improvement: Extend synchronization to cover the complete patient lifecycle, including calling, consultation initiation, completion, and no-show

handling. Ensure that timestamps and operational metrics are updated consistently.

- **Real-Time Queue Updates & Concurrent Operations:**
  - Challenge: Multiple staff members may interact with the queue simultaneously, potentially resulting in conflicting actions or outdated queue information.
  - Resolution: Integrated Server-Sent Events for queue updates and optimized Prisma transactions with longer timeouts and atomic write operations.
  - Future Improvement: Conduct additional concurrency testing for simultaneous staff actions, repeated call-next requests, and competing queue status updates. Ensure that each patient can only undergo valid queue transitions.
- **Waiting Time Estimation & Queue Accuracy:**
  - Challenge: Estimated waiting times depend on the number of patients ahead and the expected service duration. Inaccurate calculations may provide patients with misleading waiting time information.
  - Resolution: Corrected the calculation to use the number of patients ahead rather than the patient's queue position, applying a 15-minute fallback service duration.
  - Future Improvement: Review the service duration assumptions and consider incorporating actual consultation durations to improve future waiting time estimates.
- **Patient Privacy & Access Protection:**
  - Challenge: Queue information must remain accessible to the appropriate patient and staff members without exposing other patients' personal information or unauthorized records.
  - Resolution: Implemented hashed tracking tokens and an authenticated patient endpoint that only returns the requesting patient's queue information.
  - Future Improvement: Conduct further authorization and security testing for tracking tokens, patient ownership validation, and staff access to queue records across different operational states.
- **Comprehensive Queue Lifecycle Testing:**
  - Challenge: Although queue entry, status retrieval, and patient-facing tracking have been implemented, additional testing is needed to verify the complete queue lifecycle and its interaction with appointment and consultation records.
  - Resolution: Implemented queue validation rules, duplicate-entry protection, queue status retrieval, and descriptive error handling for unavailable queues.
  - Future Improvement: Expand automated testing to cover ordered patient calls, empty queues, invalid room or provider details, concurrent staff actions, and all supported queue status transitions.

What Needs to Be Done:

- **Staff Dashboard Integration:** Connect the staff dashboard to appointment records and active queue information. Display upcoming appointments, waiting patients, the currently serving ticket and assigned room, and relevant operational counts. Implement appropriate empty and no-queue states.
- **Call-Next & Queue Status APIs:** Implement the call-next behavior and associated queue status APIs. Ensure that patient calls are processed in the correct order and that room/provider information, timestamps, and queue entry lifecycle transitions are properly recorded and validated.
- **Staff Queue Operation Controls:** Develop the staff interface controls for Call Next, Start Consultation, Complete, and Mark No-Show. Implement additional skip or return actions only when explicitly approved and supported by the clinic's operational requirements.
- **Queue, Appointment & Consultation Synchronization:** Synchronize queue and appointment statuses, consultation records, timestamps, and operational metrics. Ensure that changes made through staff operations are reflected across the system while patient-facing interfaces only display information appropriate to the patient.
- **Queue Operations Testing & State Transition Validation:** Conduct comprehensive testing of ordered patient calls, empty queues, invalid room/provider details, concurrent staff actions, and all supported queue status transitions. Verify that invalid operations are rejected and that database records remain consistent.
- **Full Staff Operational Cycle Demonstration:** Demonstrate the complete clinic workflow, following a patient from queue entry through call notification, consultation, and final outcome. Document triage findings, identify remaining issues, and prepare the notification and audit event list for subsequent implementation and review.

Sprint 4 Task Reports

Sprint: CQ-S4-4.1

Member: John Alec L. Resos
Role: Tester/Developer
Task: Define active/closed queue behavior, ticket numbering, position/wait estimate, priority, timestamps, tracking token and duplicate-entry rules.

Done:

- Implemented a structured plan for the flow of queuing for both patients who booked online and walk-in.

Sprint: CQ-S4-4.2

Member: John Albert D. Masangkay / Matthew Ryan R. Sabino
Role: Technical Lead / Project Manager
Task: Implement validated walk-in and online queue entry and return tracking details such as sequential ticket number, queue position, estimated wait time, initial status, and secure tracking token.

Done:

- Implemented input validation for patient name, optional visit reason, and patient ID mapping.
- Implemented sequential ticket number generation (A-001) with initial WAITING status and hashed tracking tokens for anonymous lookups.
- Implemented dynamic queue position calculation and estimated wait time based on active queue length and fallback service duration.
- Implemented online queue entry endpoint (POST /api/v1/queues/:queueId/online) and wired the patient dashboard self-service flow so users can get a queuing number directly through the web app.
- Handled closed queue sessions (QUEUE_CLOSED) and nonexistent queues (QUEUE_NOT_FOUND) with descriptive error responses.

Sprint: CQ-S4-4.3

Member: Matthew Ryan R. Sabino
Role: Project Manager
Task: Implement queue status retrieval and integrity rules to provide real-time queue metrics while preventing duplicate active entries and unauthorized access to patient data.

Done:

- Implemented real-time queue status endpoint (GET /api/v1/queues/:queueId/status) returning currently serving ticket, counter/room designation, total waiting count, category breakdown (Online, Walk-in, Staff-generated), completed count today, and average wait time.
- Enforced duplicate-entry integrity rules (DUPLICATE_ACTIVE_ENTRY) to prevent patients from holding multiple active tickets in the same daily queue session, with Prisma P2002 conflict mapping.

- Implemented authenticated patient queue status endpoint (GET /api/v1/queues/:queueId/my-entry) to securely return active position, status, and wait estimates without leaking other patients' records or PII.
- Synchronized live counter progression and countdown timer on the Staff Console (onCallNext) and bound Patient Dashboard "Now Serving" metrics to real-time Server-Sent Events (queue:called, queue:updated).
- Optimized Prisma transactions for remote cloud database latency, raising interactive transaction timeouts to 15s and scoping locks strictly to atomic writes.

Sprint: CQ-S4-4.4

Member: Jereign B. Lim
Role: UI/UX Designer
Task: Build the patient virtual queue screen showing ticket, position, estimated wait, current serving and updated state, with empty, error and privacy safeguards. Also delivered the UI/UX redesign of the staff, patient and auth screens.

Done:

- Redesigned the staff, patient and login/register screens with a new brand palette, logo, collapsible sidebar and stat cards (styles only).
- Built the patient queue page with a ticket card and a queue summary card, plus a Queue link in the sidebar.
- Connected the summary to the queue status API with loading, closed and error states, a Refresh button, a Live/Offline indicator and live updates.
- Added a "You are not in the queue" empty state, next in line and duplicate entry banners, and the UI mapping status labels.

Sprint: CQ-S4-4.5

Member: John Alec L. Resos
Role: Tester/Developer
Task: Connect the queue UI to entry/status APIs and test numbering, wait estimates, invalid input, closed queues, duplicates and access protection.

- Focused on the patient dashboard
- **Automatic Queue Entry:** Booking an appointment automatically generates a sequential ticket (A-XXX) in WAITING status on today's queue, no manual "Join Queue" click required.
- **Accurate Wait Estimation:** Fixed the wait calculation from "position"×15"m" to "patientsAhead"×15"m" (1" patient ahead"=15" mins" , not 30" mins").

- **Unified Cancellation:** Replaced separate "Cancel Ticket" with a single "Cancel Appointment" action that simultaneously cancels both the appointment and the queue ticket in the database.
- **No-Appointment Protection:** Blocked walk-in queue entry for patients without a confirmed booking; shows "Book to Queue" instead.
- **Live UI Integration:** Connected /patient/queue to live backend APIs (getMyQueueEntry, trackByToken) and removed mock sample data.

Sprint: CQ-S4-4.6

Member: John Raphael R. Reganit
Role: Tester/Developer
Task: Collect asynchronous team updates and produce a short progress recording showing working screens, queue behavior..

Short Progress Demo Recording

https://youtu.be/QamKRlmuT2M